THE CANONICAL MAP FROM A COMPLEX INTO THE INTERNAL HOM OF ITS TOTAL TENSOR PRODUCT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Internal Hom complex of two complexes and the total tensor product, with their sign conventions. discussed in the same section, source lines 17960-18112, before the counted statement. Supplies both sides of the comparison and the signs used in the verification.
- Monoidal structure on the category of complexes and its internal Hom. the author cites Categories, Remark categories-remark-internal-hom-monoidal at proof lines 18126-18129. Explains the categorical origin of the map that the direct construction then exhibits.
- Finiteness of the components of an element of the total complex. used at proof lines 18150-18156 to show the map lands in the target. Makes the assignment well defined.

Source reference. Original lemma 0A62: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12830–12830. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0GWQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 21080–21080. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0GWJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 21074–21074. Navigation pointer to a statement the

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counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a ring. Given complexes $K^\bullet, L^\bullet$ of R-modules there is a canonical morphism*

$$K^\bullet \longrightarrow \mathrm{Hom}^\bullet(L^\bullet, \mathit{Tot}(K^\bullet \otimes_R L^\bullet))$$

of complexes of R-modules functorial in both complexes.

Proof. Via the discussion in Remark 0GWQ the existence of such a canonical map follows from Categories, Remark 0GWJ. We also give a direct construction.

Let α be an element of degree n of the right hand side. Thus

$$\alpha = (\alpha^{p,q}) \in \prod\nolimits_{p+q=n} \mathrm{Hom}_R(L^{-q}, \mathrm{Tot}^p(K^\bullet \otimes_R L^\bullet))$$

Each $\alpha^{p,q}$ is an element

$$\alpha^{p,q} = (\alpha^{r,s,q}) \in \mathrm{Hom}_R(L^{-q}, \bigoplus\nolimits_{r+s+q=n} K^r \otimes_R L^s)$$

where we think of $\alpha^{r,s,q}$ as a family of maps such that for every $x \in L^{-q}$ only a finite number of $\alpha^{r,s,q}(x)$ are nonzero. By our sign rules we get

$$\begin{aligned} \mathrm{d}(\alpha^{r,s,q}) &= \mathrm{d}_{\mathrm{Tot}(K^\bullet \otimes_R L^\bullet)} \circ \alpha^{r,s,q} - (-1)^n \alpha^{r,s,q} \circ \mathrm{d}_L \\ &= \mathrm{d}_K \circ \alpha^{r,s,q} + (-1)^r \mathrm{d}_L \circ \alpha^{r,s,q} - (-1)^n \alpha^{r,s,q} \circ \mathrm{d}_L \end{aligned}$$

Now an element $\beta \in K^n$ we send to α with $\alpha^{n,-q,q} = \beta \otimes \mathrm{id}_{L^{-q}}$ and $\alpha^{r,s,q} = 0$ if $r \neq n$. This is indeed an element as above, as for fixed q there is only one nonzero $\alpha^{r,s,q}$. The description of the differential shows this is compatible with differentials. □